



MSc/MBA Lecture Schedule – 2026 July

Semester

Program:	Intake – July 2026	Module	Code – IT5012	2 x 15 hours
	Semester - 1		Name – Mathematics for AI	
Lecturers	1. Dr. Kapila Dissanayaka	3. -----		
	2. -----	4. -----		

Week		Topic	Lecturer
1	Jul Sat 18 th	Inauguration. No Lectures.	---
	Jul Sun 19 th		
2	Jul Sat 25 th	Lecture-1: – Introduction to Mathematics for AI. – Basic algebra and functions. – Scalars and vectors.	Dr. Kapila
	Jul Sun 26 th		
3	Aug Sat 01 st	Lecture-2: – Matrices and Matrix operations. – Vector spaces, linear independence, basis, dimension.	Dr. Kapila
	Aug Sun 02 nd		
4	Aug Sat 08 th	Lecture-3: – Determinants, inverses. – Eigenvalues, eigenvectors, eigendecomposition.	Dr. Kapila
	Aug Sun 09 th		
5	Aug Sat 15 th	Lecture-4: – Linear transformations and their matrix representations.	Dr. Kapila
	Aug Sun 16 th		
6	Aug Sat 22 nd	Lecture-5: – Limits and continuity, Derivatives.	Dr. Kapila

	Aug Sun 23 rd		
7	Aug Sat 29 th	Lecture-6: – Partial derivatives, chain rule, and gradient vectors.	Dr. Kapila
	Aug Sun 30 th		
8	Sep Sat 05 th	Lecture-7: – Taylor series and approximations of functions. – Introduction to Integrals. – Multivariable calculus (multiple integrals).	Dr. Kapila
	Sep Sun 06 th		
9	Sep Sat 12 th	Lecture-8: – Gradient descent and Newton's method for optimization.	Dr. Kapila
	Sep Sun 13 th		
10	Sep Sat 19 th	Lecture-9: – Sets and set operations. – Probability theory – Conditional probability, Bayes' theorem, and Naïve Bayes Classification.	Dr. Kapila
	Sep Sun 20 th		
11	Sep Sat 26 th	Holiday (No Lecture) (Binara Full Moon Poya Day)	
	Sep Sun 27 th		
12	Oct Sat 03 rd	Lecture-10: – Discrete Random variables and Probability distributions. – Expectation and variance.	Dr. Kapila
	Oct Sun 04 th		
13	Oct Sat 10 th	Lecture-11: – Continuous Random variables and Probability distributions. – Expectation and variance.	Dr. Kapila
	Oct Sun 11 th		
14	Oct Sat 17 th	Lecture-12: – Hypothesis testing, p-values. – Type-I and Type-II errors. – Intro to parametric (Z, T, F) and non-parametric (Chi-square Goodness of fit) tests.	Dr. Kapila

	Oct Sun 18 th		
15	Oct Sat 24 th	Lecture-13: – Confidence Intervals. – Model Evaluation Metrics (R^2 , MSE, MAE, RMSE for regression; Accuracy, Precision, Recall, F1-score for classification).	
	Oct Sun 25 th	Holiday (No Lecture) (Vap Full Moon Poya Day)	
16	Oct Sat 31 st	Lecture-14: – Final Remarks. – Discussion on Final Exam.	
	Nov Sun 01 st		
		Last Day of MSc/MBA Lectures	
		FINAL EXAMS	

Assignments:

1. CA-1: A 3-page comprehensive report on the role of **Linear Algebra** in developing methods for AI, Notice on day 1 of lectures, Submit by the end of day 4 of lectures.
2. CA-2: A 3-page comprehensive report on the role of **Calculus** in developing methods for AI, Notice on day 5 of lectures, submit by the end of day 8 of lectures.
3. CA-3: A 3-page comprehensive report on the role of **Probability and Statistics** in developing methods for AI, Notice on day 8 of lectures, submit by the end of day 12 of lectures.

Recommended Text:

1. *Mathematics for Machine Learning*, M. P. Deisenroth, A. A. Faisal, C. S. Ong, 1st Ed. 2020, Cambridge University Press.
2. *Schaum's Outline of Linear Algebra*, 6th Ed. 2017, McGraw Hill.
3. *Calculus*, J. Stewart, 8th Ed. 2015, Cengage Learning.
4. *Discrete Mathematics and It's applications*, K. H. Rosen, 7th Ed. 2011, McGraw Hill.
5. *Introduction to Mathematical Statistics*, R. Hogg, J. McKean, A. Craig, 8th Ed. Pearson, 2018.